Short Answer :-

Q.1) Why is it necessary to separate oxygenated and deoxygenated blood in mammals and birds?

Ans:) The mammals and birds continuously need more energy to maintain their body temperature. Therefore, to provide more energy it is necessary to separate oxygenated and deoxygenated blood. So that the organism like mammals and birds can get more oxygen with high efficiency to maintain their body temperature.

Q.2) What is the role of saliva in the digestion of food?

Ans:) Saliva is secreted by the salivary glands located under the tongue. It makes the food soft and smooth for easy swallowing. It contains a digestive enzym called salivary amylase, which breaksdown starch into sugar. In this way the saliva helps in digestion of food.

Q.3) Describe double circulation in human beings Why is it necessary?

Ans:) "In each of the cycle the blood

enters in the heart two times, this process is called double circulation."
In this process oxygenated blood eters in our heart from lungs while second time the deoxygenated blood from other parts of the body enters in our heart.

The double circulation in our body is necessary because it separates the oxygenated and deoxygenated blood from mixing each other due to which we receive more energy efficiently.

Q4) Compare the functioning of alveoli in the lungs and nephrons in the kidneys with respect to their structure and functioning.

Ans⇒ Alveoli are tiny balloon like structures present inside the lungs with the thick walls and contains an extensive network of blood capilliaries while nephrons are tubular structure present inside the kidney made of glomerulus, bowmans capsule a cup shaped structure and contains a cluster of thin walled capillaries.

The main function of alveoli is to exchange gases like carbon dioxide (CO_2)

and oxygen (O_2) while the main function of nephrons is to filter urine and to reabsorb selectively the useful necessary substances like glucose, amino acids, water, etc. from the urine.

Long Answer:

Q.5) Explain the process of inhalation (inspiration) and exhalation (expiration).

Or.

Explain the process of breathing in man.

Or.

In how many steps does breathing occur? Explain them.

Ans.) The process of breathing: Breathing is a part of respiration and is called external respiration which occurs in two steps -

1) Inhalation (Inspiration).
2) Exhalation (Expiration).

1) Inhalation (Inspiration): When diaphragm expands, the volume of lungs increases which suck the oxygenated air from atmosphere through nasal cavity. This air comes in contact with blood by virtue of which haemoglobin present in blood absorbs the oxygen. Blood release carbon dioxide gas and

water vapour.

2) Exhalation (Expiration) ⇒ When diaphragm contracts, the pressure is exerted on lungs which pushes out air containing water vapour and carbon dioxide which diffuses in atmosphere.

Q.6) Describe the process of urine formation in kidneys.

Ans.) The process of urine formation in kidneys:
The blood is supplied to kidney by renal arteries through arterioles which enter the glomerulus situated in Bowman's capsule. Here the filtration of the blood takes place due to which the soluble substances like glucose, salts, urea and uric acid etc. come out as filtrate by ultrafiltration. The filtrate now passes through very fine capillaries where useful substances like glucose and other salts are reabsorbed and are transported to the blood through renal veins. The, residual filtrate contains water, urea, uric acid, some salts, pigments, etc. This residual solution is called urine. Thus the process of urine formation takes place in the kidneys.